

Pranshu Gupta

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PROFESSIONAL EXPERIENCE

- **Microsoft AI** / *Member of Technical Staff* / Redmond, WA Aug 2025 – Present
 - **Principal technical authority** for LLM inference on Copilot, shaping architectural decisions across model integration, safety infrastructure, and inference reliability.
 - Engineered the inference pipeline for **GPT-5.1 instant and reasoning models** on Copilot at launch, with meta-prompting for personality alignment, delivering a **30% improvement in personality evals** for the reasoning path.
 - Implemented a new **constrained sampling approach** for tool calls in GPT-5.1 reasoning models, verified through token-level analysis, preventing a production regression in tool call reliability at Copilot launch.
 - Scoped, designed, and built **GPU-based safety classifier** support in Chat Completions API: non-streaming endpoints returning classification logits for client-side threshold scoring across input and output streams.
 - Designed and shipped an **incremental CoT summarization** system: buffers streaming reasoning chunks and progressively surfaces summaries in the UI, keeping it responsive while protecting raw model reasoning as IP.
 - Reduced inference costs by eliminating non-essential classifiers (**reducing GPU footprint by 20%**) and resolved a token overhead inefficiency in Chat Completions API (**reducing input tokens by 25%** and **TTFT by 30ms**).
 - Defined and instrumented LLM API **traffic classification** dimensions (product, client app, inference intent, license tier) across MAI and M365 Copilot; built Grafana dashboards informing capacity planning and cost attribution across both orgs.
- **Microsoft Azure AI** / *Senior Software Engineer* / Redmond, WA Jan 2025 – Aug 2025
 - Designed and built a distributed **token-based rate limiter** using Tiktoken for precise token counting, replacing a length-based estimator with 219% average error; implemented pub-sub quota state synchronization across instances.
 - Designed and built the inference pipeline for an intelligent **model router**: injecting per-request model selection based on cost/complexity signals before each LLM call, reducing per-inference cost by **55%**.
- **Microsoft Azure Core** / *Senior Software Engineer* / Redmond, WA Mar 2024 – Jan 2025
 - Served as **technical area owner for resource lifecycle management** in Azure Resource Manager.
 - Mentored a new-grad engineer through weekly 1:1s; they now independently own a critical component in control plane.
 - Designed and shipped a multi-bucket **token-based background job throttling** system with retry-storm protection, preventing noisy-neighbor throttling in Azure Control Plane background-job infrastructure.
 - Single-handedly designed and shipped **async operation URI signing**; ran office hours with resource provider teams to guide API surface changes, adopted by **242 service teams** with **220+ enforcing**.
- **Microsoft Azure Core** / *Software Engineer II* / Redmond, WA Jun 2020 – Mar 2024
 - Designed and shipped **async operation callbacks** replacing polling, cutting control plane polling traffic by **94%**.
 - Led **Azure region teardown** end-to-end: halted new resource creation, resolved subscription/resource group hierarchies, and migrated live resources to healthy regions with zero data loss.
 - Designed and built a **health-driven deployment pipeline** for ARM control-plane manifests: staged multi-region rollouts with post-step RP health checks that halt or rollback on degradation.
- **Microsoft Core Services** / *Software Engineer* / Hyderabad, India Jun 2017 – Aug 2019
 - Modernized a data enrichment platform to Azure Function Apps (**\$4K/month** savings); integrated Cognitive Services for data standardization, and built a bulk-processing pipeline with **40–50x throughput** improvement.

TECHNICAL SKILLS

- **AI/ML & Inference:** LLM inference/serving, prompt engineering, model evaluations, safety orchestration, RAG
- **Languages & Systems:** Python, .NET, C++, KQL, JavaScript, HTML, CSS, Kubernetes, Redis, Grafana
- **Cloud & Infrastructure:** Azure (Cosmos DB, ARM Templates, Service Bus, DevOps), AWS (Bedrock, S3, ECR)

SELECTED PROJECTS

- **AI Lab Assistant** (TurnTheBus nonprofit, 2025–Present): Architected and built end-to-end on **AWS** using **RAG** over physics lab manuals to guide high school students through virtual physics experiments.

PUBLICATIONS & WRITING

- *Runtime Optimizations for LLMs* (2026). Production notes on inference optimizations for latency/cost-efficient LLM serving.
- *The Observer Effect on Social Media* — ACM CHI (2024). Causal inference and time series study on longitudinal Facebook data from 300+ participants to quantify how monitoring awareness alters online behavior.
- *Mental Health Impact of Active-Shooter Drills* — Nature (2021). Applied ML classifiers and interrupted time series analysis to 54M social media posts across 114 schools to measure psychological and civic impact.

EDUCATION

- **Georgia Institute of Technology** / *M.S. Computer Science (Machine Learning)* / Atlanta, GA May 2020
- **Indian Institute of Technology, Kanpur** / *B.Tech, C.S. and Eng.* / Kanpur, India May 2017